Connections

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1 Motivation

Everything so far (the set, the topology, the topological and smooth manifold structure, the bundles) is structure that has to be supplied by hand before physics can start, because there are no equations yet that determine it. Today we add another structure on a smooth manifold: a **connection**, also called a *covariant derivative* (strictly the two notions differ slightly, but for us they coincide; "affine connection" and "linear connection" are further names). In general relativity the connection will be determined by the Einstein equations: they determine a metric, a still more special structure, which in turn determines a connection. Here we remain purely mathematical.

From now on we work mostly with *vector fields* and denote them by capital letters X, Y, Z. (The bundle construction of the previous lecture is needed to speak precisely about smooth vector fields, but for intuition "a vector at each point" is enough.)

2 The goal: directional derivatives of tensor fields

A vector field X provides a directional derivative of smooth functions $f \in C^\infty(M)$:

$$X(f) \;=\; (\mathrm{d}f)(X) \;=:\; \nabla_X f.$$

These are three notations for the same function. The first has X acting on f; the second has $\mathrm{d}f$ acting on X; the third emphasizes that we are taking a directional derivative of f in the direction X. The new symbol is worthwhile for the following reason. $X(f)$ can only eat functions, i.e. $(0,0)$ tensor fields. We want to extend the directional derivative to act on arbitrary (p,q) tensor fields, for instance to differentiate a vector field in a given direction. This extension does *not* come for free on a smooth manifold. Even if we demand every property a directional derivative should have, extra structure is needed. The symbol ∇ will stand for exactly this additional structure, and on functions it reduces to the old action $\nabla_X f := X(f)$.

3 Definition of a connection

We write a wish list of properties and then see how much freedom is left.

Definition 1 (Connection)**.** A *connection* on a smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is a map ∇ that takes a vector field X and a (p,q) tensor field T to a (p,q) tensor field $\nabla_X T$, such that:

(i) **Extension:** on $(0,0)$ tensor fields (functions), $\nabla_X f = X(f)$.

(ii) **Additivity in T:** for T, S both of type (p,q), $\nabla_X(T+S) = \nabla_X T + \nabla_X S$.

(iii) **Leibniz rule:** for a $(1,1)$ tensor field T, a covector field ω and a vector field Y,

$$\nabla_X\big(T(\omega, Y)\big) = (\nabla_X T)(\omega, Y) + T(\nabla_X \omega, Y) + T(\omega, \nabla_X Y),$$

and analogously for any (p,q) tensor field (one extra term for each additional slot).

(iv) $C^\infty(M)$**-linearity in the lower slot:** for $f \in C^\infty(M)$,

$$\nabla_{fX+Z}\, T = f\, \nabla_X T + \nabla_Z T.$$

A *manifold with connection* is a quadruple $(M, \mathcal{O}, \mathcal{A}, \nabla)$.

Notes:

- The left side of (iii) is already a smooth function, so it equals ∇_X of that function; this is how (i) and (iii) interlock. On the problem sheet one checks that all four properties are consistent (in particular that (iii) reduces correctly for $p = q = 0$).
- The Leibniz rule can equivalently be written with the tensor product: for a (p, q) tensor field T and an (r, s) tensor field S, $T \otimes S$ is the $(p + r, q + s)$ tensor field defined by evaluating T on the first p covector fields and first q vector fields and S on the rest, multiplying the results (in $\mathbb{R}$, or in $C^\infty(M)$ for fields); then

$$\nabla_X(T \otimes S) = \nabla_X T \otimes S + T \otimes \nabla_X S.$$

- Property (iv) says that $\nabla_X T$ is $C^\infty(M)$-linear in X. It follows that $(\nabla_X T)(p)$ depends only on the value of X at the point p, not on how X behaves in a neighborhood. So one could feed in a single vector $X_p \in T_pM$ and get a tensor at p; this is the same logic as a tensor field being C^∞-linear in every slot. (By contrast, the Lie derivative needs a whole vector field.)
- The dependence on T is only additive, not C^∞-linear: a derivative obeys a product rule, $\nabla_X(fg) = X(f)\, g + f\, X(g)$, so $\nabla_X(fT) \neq f \nabla_X T$.
- ∇ without the subscript is the generalization of d. It needs a vector field fed in before it can act on a function.

4 How much freedom is left? Connection coefficients

To see what data a connection contains, go to a chart (U, x) and act on a vector field. Expand $X = X^i\, \partial/\partial x^i$ and $Y = Y^m\, \partial/\partial x^m$ with component functions $X^i, Y^m \in C^\infty(M)$ (here $\partial/\partial x^i$ denotes the chart-induced vector field on U). By additivity and C^∞-linearity in X, and the Leibniz rule for $Y^m \cdot \partial_m$:

$$\nabla_X Y = X^i\, \nabla_{\partial_i}\big(Y^m \partial_m\big) = X^i \Big(\partial_i(Y^m)\, \partial_m + Y^m\, \nabla_{\partial_i} \partial_m\Big).$$

(Here $\partial_i(Y^m)$ is the vector field acting on a function.) The last object $\nabla_{\partial_i}\partial_m$ is again a vector field, so it can be expanded in the chart-induced basis, with coefficient functions that we have to name.

Definition 2 (Connection coefficient functions)**.** Let $(M, \mathcal{O}, \mathcal{A}, \nabla)$ be a manifold with connection and $(U, x) \in \mathcal{A}$. The *connection coefficient functions* (or *Christoffel symbols* in the metric context) of ∇ with respect to (U, x) are the $(\dim M)^3$ functions ${\Gamma^i}_{jk} : U \to \mathbb{R}$,

$${\Gamma^i}_{jk} \;:=\; \mathrm{d}x^i \left(\nabla_{\partial/\partial x^k}\, \frac{\partial}{\partial x^j} \right).$$

Warning (Convention). The *last* lower index is the direction of differentiation (the slot of the subscript on ∇). Many authors use another ordering, so always check the convention. Equivalently, $\nabla_{\partial_k}\partial_j = {\Gamma^i}_{jk}\, \partial_i$.

The Γ's are chart-dependent functions, but nevertheless functions on the manifold (on the chart domain); we regard them as functions of a point of M, not of a point of the chart image, which avoids carrying x^{-1} around. Examples of other chart-dependent functions on M are the coordinate functions x^i themselves. If necessary one writes a subscript, ${\Gamma_{(x)}}^i{}_{jk}$, to indicate the chart; this is advisable when several charts are in play.

4.1 Action on vector fields

$$\boxed{(\nabla_X Y)^i = X(Y^i) + \Gamma^i{}_{jm}\, X^m\, Y^j} \tag{1}$$

Here $X(Y^i) = X^m \partial_m(Y^i)$ is the vector field acting on the i-th component *function*, which is why the index i must sit inside the brackets. The correction term is linear in X, with the direction index m of Γ contracted with X^m; the remaining indices are fixed by the requirement that the result has one free upper index i.

4.2 Action on covector fields

Apply ∇_{∂_m} to the function $\mathrm{d}x^i(\partial_j) = \delta^i{}_j$ (a constant function, so its derivative vanishes) and use the Leibniz rule:

$$0 = (\nabla_{\partial_m} \mathrm{d}x^i)(\partial_j) + \mathrm{d}x^i\big(\nabla_{\partial_m} \partial_j\big) = (\nabla_{\partial_m} \mathrm{d}x^i)(\partial_j) + \Gamma^i{}_{jm}.$$

Hence $\nabla_{\partial_m} \mathrm{d}x^i = -\Gamma^i{}_{jm}\, \mathrm{d}x^j$: the same coefficient functions appear, with a *minus* sign, so no new data is needed. For a covector field $\omega = \omega_i\, \mathrm{d}x^i$:

$$\boxed{(\nabla_X \omega)_i = X(\omega_i) - \Gamma^j{}_{im}\, X^m\, \omega_j} \tag{2}$$

4.3 General tensor fields

Repeating the argument with the Leibniz rule: *act with X naively on the component functions, then add one Γ-term for each upper index (with a plus sign) and one for each lower index (with a minus sign)*, keeping all other indices of T untouched and using dummy indices. For example, for a $(1,2)$ tensor field,

$$(\nabla_X T)^i{}_{jk} = X\big(T^i{}_{jk}\big) + \Gamma^i{}_{sm} X^m\, T^s{}_{jk} - \Gamma^s{}_{jm} X^m\, T^i{}_{sk} - \Gamma^s{}_{km} X^m\, T^i{}_{js}.$$

The direction index m of Γ is always contracted with X; the upper index of Γ is the one of the term it replaces (for an upper index of T) or the dummy index contracted with T (for a lower index of T).

Theorem 1 (Upshot)**.** *On a chart domain, the* $(\dim M)^3$ *functions* $\Gamma^i{}_{jk}$ *determine the action of* ∇ *on* every *tensor field, whatever its rank. A connection is therefore additional structure that must be supplied; the freedom left by the four axioms is precisely this choice of functions (subject to the compatibility between charts found below).*

These index rules are used constantly in general relativity and are worth committing to memory.

4.4 Example: Euclidean connection and others

On $M = \mathbb{R}^3$ with the chart $(\mathbb{R}^3, \mathrm{id})$ one may *choose* all $\Gamma^i{}_{jk} = 0$. This defines the Euclidean connection. It is a choice: the same smooth manifold $\mathbb{R}^3$ can be equipped with a different connection (e.g. a "hyperbolic" one, whose shape is that of a saddle, though the manifold still extends infinitely in all directions). How the connection relates to curvature and shape is the topic of the next lecture. And even for the Euclidean connection, in another chart such as polar coordinates the coefficients are *not* all zero.

Example 1 (Divergence)**.** For a vector field X on a manifold with connection, the divergence is the trace of ∇X:

$$\operatorname{div} X := \mathrm{d}x^i\big(\nabla_{\partial_i} X\big) = \partial_i X^i + \Gamma^i{}_{ji} X^j.$$

It is chart independent (a trace of a $(1,1)$ tensor field). In a chart where the Γ vanish it reduces to the familiar sum of partial derivatives. In polar, spherical, cylindrical or other coordinates the additional terms in every formula collection are nothing but the Γ terms of that chart.

5 Change of chart

The Γ's were defined chart by chart. Charts overlap, so a choice of Γ's on one chart constrains the choice on another.

Let (U, x) and (V, y) be charts with $U \cap V \neq \emptyset$. Write $\Gamma_{(x)}$ and $\Gamma_{(y)}$ for the coefficients in the two charts. Starting from ${\Gamma_{(y)}}^i{}_{jk} = \mathrm{d}y^i(\nabla_{\partial/\partial y^k}\partial/\partial y^j)$, expanding $\partial/\partial y^j = \frac{\partial x^s}{\partial y^j}\partial/\partial x^s$, using the chain rule, the Leibniz rule and C^∞-linearity in the lower slot gives

$$\boxed{{\Gamma_{(y)}}^i{}_{jk} = \frac{\partial y^i}{\partial x^q}\,\frac{\partial x^s}{\partial y^j}\,\frac{\partial x^p}{\partial y^k}\,{\Gamma_{(x)}}^q{}_{sp} \;+\; \frac{\partial y^i}{\partial x^q}\,\frac{\partial^2 x^q}{\partial y^j\,\partial y^k}} \tag{3}$$

(The lecture derives this on the board with some index relabelling; the form above is the standard one in the convention of this section.)

Reading the formula.

- The first term is the transformation law of a $(1,2)$ tensor's components. The second term is *inhomogeneous*: it does not contain Γ. Hence the connection coefficient functions are **not** the components of a tensor.
- The second derivative of x is taken with respect to *one* coordinate system y, so it is symmetric in j, k (Schwarz). One cannot exchange derivatives with respect to different coordinate sets; this is a common mistake.
- For tensors: if the components vanish in one chart they vanish in every chart. For Γ this fails: Γ can vanish in one chart (even throughout the manifold) and be non-zero in another, whenever the chart transition has non-trivial second derivatives. Under linear changes of coordinates the second term does not contribute, so one does not notice it; this is one reason why in special relativity it seems natural to restrict to Lorentz transformations in position space (as opposed to the tangent spaces, where they are linear anyway). Passing to polar coordinates, by contrast, produces non-zero Γ from zero.
- Compatibility: if $\Gamma_{(x)}$ is chosen on U and $\Gamma_{(y)}$ on V, they must obey (3) on $U \cap V$, otherwise ∇ would be ill-defined on the overlap. So the choices cannot be made independently on all charts.
- The transformation law contains the second term only in the symmetric combination in (j, k). Hence the *antisymmetric* part ${\Gamma^i}_{[jk]}$ transforms homogeneously and is a tensor. It is the *torsion* (to be treated in the next lecture); a connection with vanishing torsion is called *torsion-free*.

6 Normal coordinates

Theorem 2 (Normal coordinates at a point)**.** *Let ∇ be a connection on M and $p \in M$. Then there is a chart (U, x) with $p \in U$ such that the symmetric part of the connection coefficients vanishes at p:*

$${\Gamma_{(x)}}^i{}_{(jk)}(p) = 0.$$

In particular, for a torsion-free *connection one can always find a chart in which* all ${\Gamma^i}_{jk}$ *vanish at p.*

Constructive proof. Start with any chart (V, y) containing p, shifted so that $y(p) = 0$ (a translation of the chart does not change the structure of what follows). Its coefficients ${\Gamma_{(y)}}^i{}_{jk}$ are generically non-zero. Define a new chart x by the chart transition map

$$x^i \circ y^{-1}(\alpha) \;=\; \alpha^i + \tfrac{1}{2}\,{\Gamma_{(y)}}^i{}_{(jk)}(p)\;\alpha^j\alpha^k,$$

where $\Gamma_{(y)}(p)$ are just fixed numbers. (This is a valid chart near p, because at $\alpha = 0$ the Jacobian is the identity, which is invertible.) At $\alpha = 0$ we have $\partial x^i/\partial y^j = \delta^i{}_j$ and $\partial^2 x^i/\partial y^j \partial y^k = \Gamma_{(y)}{}^i{}_{(jk)}(p)$. Since the Jacobian of the transition is the identity at p, the second derivative of the inverse transition is minus that of the original, $\partial^2 y^q/\partial x^j \partial x^k|_p = -\Gamma_{(y)}{}^q{}_{(jk)}(p)$, and applying the transformation law (3) with the roles of the charts exchanged gives at p

$$\Gamma_{(x)}{}^i{}_{jk}(p) = \Gamma_{(y)}{}^i{}_{jk}(p) - \Gamma_{(y)}{}^i{}_{(jk)}(p) = \Gamma_{(y)}{}^i{}_{[jk]}(p).$$

The symmetric part has been removed, and only the antisymmetric part (the torsion) remains. For a torsion-free connection that part is zero as well. □

Remark. The lecture runs this computation with the transformation law in the other direction and fixes the sign ("plus or minus, I will decide in a second") by requiring the cancellation; the construction above is arranged so that the cancellation can be checked directly in convention (3). The lecturer also initially claimed the full Γ vanishes and then corrected it to the symmetric part.

Terminology. Such a chart is called a *normal coordinate chart* of ∇ at p. (These are what one might loosely call "Cartesian" coordinates, but angles and lengths require a metric, which we do not have yet.)

Only at a point. The construction uses the fixed numbers $\Gamma_{(y)}(p)$ at one chosen point. Choosing the coefficients to depend on position would destroy the form of the second derivative used in the proof. In general one *cannot* find a chart in which Γ vanishes throughout a whole neighborhood of p. The obstruction is *curvature*, to be defined in the next lecture. There it is shown that a chart in which all Γ vanish on a neighborhood exists if and only if both torsion and curvature of the connection vanish there (the connection is "flat" and torsion-free). The statement that it is impossible by this particular construction does not by itself prove that no other construction works; that is the content of the converse theorem. (Later one can also arrange vanishing along straight lines, but not in a neighborhood.)

Summary

- A **connection** ∇ extends the directional derivative $\nabla_X f = X(f)$ to all tensor fields, subject to: extension on functions, additivity, the Leibniz rule, and $C^\infty(M)$-linearity in the direction X. It is extra structure on a smooth manifold.
- In a chart it is encoded by $(\dim M)^3$ functions $\Gamma^i{}_{jk} = \mathrm{d}x^i(\nabla_{\partial_k}\partial_j)$, which fix the action on every tensor field: $(\nabla_X Y)^i = X(Y^i) + \Gamma^i{}_{jm} X^m Y^j$, $(\nabla_X \omega)_i = X(\omega_i) - \Gamma^j{}_{im} X^m \omega_j$, and in general $+$ for each upper and $-$ for each lower index.
- Under a change of chart Γ transforms like a $(1,2)$ tensor plus an inhomogeneous term involving second derivatives of the transition map. Γ is therefore not a tensor, and the choices on overlapping charts are constrained. Its antisymmetric part is a tensor (torsion).
- At any point there are **normal coordinates** in which the symmetric part of Γ vanishes (all of Γ if torsion-free), but generally not on a whole neighborhood: curvature is the obstruction.